

The diagram illustrates a differential pair circuit with current mirrors and a tail current source. The circuit is powered by v_{dd} and v_{ss} .

- Input Stage:** Two NMOS transistors, X_{inp} (LVT, $3/0.15$) and X_{inn} (LVT, $3/0.15$), are connected in a differential pair. Their gates are driven by v_{inp} and v_{inn} respectively. The sources of X_{inp} and X_{inn} are connected to a common tail node labeled **tail**.
- Current Mirrors:**
 - PMOS transistors X_{pn} ($2/0.15$) and X_{pp} ($2/0.15$) are connected to v_{dd} . Their gates are tied together and connected to the drain of X_{inp} (node **qn**) and the drain of X_{inn} (node **xn**) via a cross-connection. Specifically, the gate of X_{pn} is connected to node **xn**, and the gate of X_{pp} is connected to node **qn**.
 - NMOS transistors X_{ln} ($1/0.15$) and X_{lp} ($1/0.15$) are connected to v_{ss} . Their gates are tied together and connected to the drain of X_{inp} (node **qn**) and the drain of X_{inn} (node **xn**) via a cross-connection. Specifically, the gate of X_{ln} is connected to node **xn**, and the gate of X_{lp} is connected to node **qn**.
- Tail Current Source:** A PMOS transistor X_{tail} ($7/0.15$) is connected between the tail node and v_{ss} . Its gate is driven by clk .

node = xp, xn, qp, qn

$X_s: W = 1, 2, 4, 8; L = 0.15$

W/L in μm . All nMOS bulks: vss; pMOS bulks: vdd. Bulk wires omitted.